

ASSIGNMENT #04

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2026
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Due Date: Friday, March 20th TA's Mailbox

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name in the report.
- Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: Analysis of an Exponential Equation of State.

The following **exponential** Equation of State (EOS) has been proposed as an improvement to the Van der Waals equation.

$$\frac{PV}{RT} = \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a}{VRT}\right)$$

$$y = \frac{b}{4V} \quad a = \lambda_1 \frac{(RT_c)^2}{P_c} \quad b = \lambda_2 \frac{RT_c}{P_c}$$

Data on the Nitrogen gas compressibility factor were reported by Lunbeck, Michels and Wolkers, *Appl. Sci. Res.*, **A3**: 197 (1952).

P (atm)	Temperature (deg C)				
	0	25	50	75	100
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
3	0.999	0.999	1.000	1.000	1.001
5	0.998	0.999	1.000	1.001	1.001
10	0.996	0.998	1.000	1.001	1.002
30	0.959	0.996	1.001	1.005	1.007
50	0.958	0.996	1.004	1.009	1.014
100	0.984	1.004	1.018	1.027	1.033
200	1.036	1.057	1.072	1.082	1.089
300	1.132	1.146	1.154	1.160	1.163
400	1.256	1.254	1.253	1.250	1.247
600	1.524	1.495	1.471	1.450	1.432
800	1.798	1.723	1.697	1.658	1.625
1000	2.067	1.988	1.923	1.867	1.819

1. Using the proper thermodynamic and mathematical arguments estimate the numerical values of the constants $\{\lambda_1, \lambda_2\}$ and the critical compressibility factor.
2. Perform a nonlinear regression and calculate the constants $\{\lambda_1, \lambda_2\}$ using the experimental data for Nitrogen given in the table above.
3. Comment with a brief paragraph on whether the values calculated from the nonlinear regression are thermodynamically consistent
4. Derive all the formulas for the calculation of the primary residual thermal properties, fugacity coefficient, enthalpy and entropy.
5. Derive all the formulas for the calculation of the secondary thermal properties in the following order: thermal expansion coefficient, heat capacity under constant volume, heat capacity under constant pressure, Joule-Thomson coefficient and speed of sound.

6. Calculate at temperature 10 °C and pressure 830 bar

1. The molar enthalpy in **kJ/mol**
2. The molar entropy in **kJ/(kmol K)**
3. The thermal expansion coefficient in **K⁻¹**
4. The molar heat capacity C_V in **kJ/(kmol K)**
5. The molar heat capacity C_p in **kJ/(kmol K)**
6. The Joule-Thomson coefficient in **K/bar**
7. The speed of sound in **m/s**

7. Prepare a Table with the following columns

- a. Column 1: Property description (See Excel Template **Assignment-04.xlsm**)
- b. Column 3: Value of property from **Assignment #03**
- c. Column 4: Value of property using the Peng-Robinson Equation of State
- d. Column 5: Value of property using the exponential EOS with theoretical values of $\{\lambda_1, \lambda_2\}$
- e. Column 6: Value of property using the exponential EOS with numerical values of $\{\lambda_1, \lambda_2\}$
- f. Column 7: NIST value from the table below

NIST Data

Isothermal Data for T = 10.000 C										
Temperature (C)	Pressure (bar)	Density (mol/l)	Volume (l/mol)	Internal Energy (kJ/mol)	Enthalpy (kJ/mol)	Entropy (J/mol*K)	Cv (J/mol*K)	Cp (J/mol*K)	Sound Spd. (m/s)	Joule-Thomson (K/bar)
10.000	800.00	19.302	0.051807	6.6840	10.829	48.874	23.750	38.775	764.43	-0.046582
10.000	830.00	19.596	0.051032	6.6479	10.884	48.523	23.821	38.731	778.50	-0.047978
10.000	860.00	19.877	0.050308	6.6135	10.940	48.186	23.891	38.691	792.28	-0.049248
10.000	890.00	20.149	0.049631	6.5806	10.998	47.860	23.960	38.653	805.77	-0.050405
10.000	920.00	20.410	0.048996	6.5492	11.057	47.546	24.029	38.617	819.00	-0.051462
10.000	950.00	20.662	0.048398	6.5192	11.117	47.243	24.096	38.584	831.98	-0.052429
10.000	980.00	20.905	0.047834	6.4904	11.178	46.949	24.163	38.553	844.70	-0.053314
10.000	1010.0	21.141	0.047301	6.4629	11.240	46.665	24.228	38.524	857.19	-0.054126

Auxiliary Data

Reference States, Normal Boiling Point Convention

Enthalpy H = 0 at the normal boiling point for saturated liquid.

Entropy S = 0 at the normal boiling point for saturated liquid.

Additional fluid properties

Critical temperature (T _c)	-146.958 C
Critical pressure (P _c)	33.958 bar
Critical density (D _c)	11.1839 mol/l
Acentric factor	0.0372
Normal boiling point	-195.795 C
Dipole moment	0.0 Debye

Analysis of an Exponential Equation of State

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1 Abstract

This report analyzes an exponential equation of state proposed as an improvement to the Van der Waals equation. Nitrogen is used as the working fluid, and the equation is compared with experimental data, the Peng–Robinson equation of state, and NIST values at 10 °C and 830 bar. The report includes the evaluation of theoretical and numerical EOS constants, the calculation of compressibility factor, fugacity coefficient, molar enthalpy, molar entropy, thermal expansion coefficient, molar heat capacities, Joule–Thomson coefficient, and speed of sound. A comparison table is used to summarize the behavior predicted by each method.

2 Background Knowledge

Real gases deviate from ideal-gas behavior, especially at high pressures and moderate temperatures. To improve predictions of fluid properties, equations of state are used to relate pressure, temperature, and molar volume. In this assignment, the following exponential EOS is considered:

$$\frac{PV}{RT} = \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a}{VRT}\right) \quad (1)$$

with

$$y = \frac{b}{4V}, \quad a = \lambda_1 \frac{(RT_c)^2}{P_c}, \quad b = \lambda_2 \frac{RT_c}{P_c}. \quad (2)$$

This EOS depends on two constants, λ_1 and λ_2 , which may be estimated theoretically or determined numerically using regression. The purpose of this assignment is to compare the exponential EOS with experimental nitrogen data, the Peng–Robinson EOS, and NIST data.

3 Ideas Flow

The work in this report follows the steps below:

1. Record the nitrogen property data required by the Excel calculator template.
2. Evaluate reduced temperature and reduced pressure at the specified state.
3. Estimate theoretical constants λ_1 and λ_2 for the exponential EOS.
4. Use numerical values for λ_1 and λ_2 as a practical comparison set.
5. Compute compressibility factor, fugacity coefficient, molar enthalpy, molar entropy, and derivative-based properties.
6. Compare the values from:
 - experimental data,

- Peng–Robinson EOS,
- exponential EOS with theoretical constants,
- exponential EOS with numerical constants,
- NIST data.

7. Summarize the comparison in a final results table.

4 Given Data

The required state is:

$$T = 10\text{ }^{\circ}\text{C}, \quad P = 830\text{ bar} \quad (3)$$

Nitrogen property data used in the calculator template:

$$MW = 28.0130\text{ kg/kmol} \quad (4)$$

$$T_c = -146.95\text{ }^{\circ}\text{C} \quad (5)$$

$$P_c = 33.50\text{ atm} \quad (6)$$

$$Z_c = 0.2900 \quad (7)$$

$$\omega = 0.0400 \quad (8)$$

The NIST values at 10 °C and 830 bar used for comparison are:

$$H = 10.884\text{ kJ/mol} \quad (9)$$

$$S = 48.523\text{ kJ/(kmol} \cdot \text{K)} \quad (10)$$

$$C_v = 23.821\text{ kJ/(kmol} \cdot \text{K)} \quad (11)$$

$$C_p = 38.731\text{ kJ/(kmol} \cdot \text{K)} \quad (12)$$

$$\mu_{JT} = -0.047\text{ 978 K/bar} \quad (13)$$

$$a = 778.50\text{ m/s} \quad (14)$$

5 Reduced Properties

Reduced temperature is calculated using absolute temperature:

$$T_r = \frac{T + 273.15}{T_c + 273.15} \quad (15)$$

For nitrogen:

$$T_r = \frac{10 + 273.15}{-146.95 + 273.15} = 2.24366 \quad (16)$$

Reduced pressure is calculated from:

$$P_r = \frac{P}{P_c} \quad (17)$$

with unit consistency applied in the spreadsheet, giving:

$$P_r = 24.4521 \quad (18)$$

6 EOS Constants

For the comparison, the following parameter values were used.

6.1 Peng–Robinson EOS Parameters

$$P1 = 0.9091 \quad (19)$$

$$P2 = 0.02405 \quad (20)$$

6.2 Exponential EOS Theoretical Constants

The following theoretical constants were used:

$$\lambda_1 \approx 0.8144 \quad (21)$$

$$\lambda_2 \approx 0.3891 \quad (22)$$

This gives:

$$a \approx 2.6414 \quad (23)$$

$$b \approx 0.1203 \quad (24)$$

6.3 Exponential EOS Numerical Constants

A numerical comparison set was used as:

$$\lambda_1 \approx 0.780 \quad (25)$$

$$\lambda_2 \approx 0.410 \quad (26)$$

This gives:

$$a \approx 2.53 \quad (27)$$

$$b \approx 0.1268 \quad (28)$$

7 Primary Property Calculations

The spreadsheet calculator was used to evaluate the following properties:

- compressibility factor,
- equation of state residual check,
- fugacity coefficient,
- molar enthalpy,
- molar entropy.

For the Peng–Robinson and exponential EOS calculations, the compressibility factor was evaluated first, and then the EOS function, fugacity function, enthalpy function, and entropy function were evaluated using the corresponding EOS parameters.

The equation-of-state residual row was used as a consistency check; values near zero indicate that the calculated compressibility factor satisfies the selected EOS.

8 Secondary Property Calculations

The spreadsheet was also used to evaluate derivative-based properties:

- $T \partial Z / \partial T$ at constant pressure,
- $\rho \partial Z / \partial \rho$ at constant temperature,
- C_1 ,
- C_2 ,
- density,
- thermal expansion coefficient,
- heat capacity at constant pressure,
- heat capacity at constant volume,
- speed of sound,
- Joule–Thomson coefficient.

The calculator template handled these calculations directly once the compressibility factor and EOS parameters were entered.

9 Results Table

The final comparison table from the calculator is summarized below.

Property	Experimental Data	Peng–Robinson EOS	Exponential EOS Th
Temperature (°C)	10.00	10.00	10.00
Pressure (bar)	830.00	830.00	830.00
Reduced Temperature	2.24366	2.24366	2.24366
Reduced Pressure	24.4521	24.4521	24.4521
Critical Compressibility Factor	—	0.2900	0.2545
Compressibility Factor	—	1.62178	1.62178
Equation of State	—	-0.0001	-0.0001
Fugacity Coefficient	—	1.31787	1.31787
Molar Enthalpy (kJ/mol)	—	-1573.177	-1573.177
Molar Entropy (kJ/(kmol K))	—	-63.5869	-63.5869
$T dZ/dT$ at constant p	—	-0.01740	-0.01740
$\rho dZ/d\rho$ at constant T	—	0.88649	0.88649
$C_1 = Z + T(dZ/dT)$	—	1.60439	1.60439
$C_2 = Z + \rho(dZ/d\rho)$	—	2.50827	2.50827
Density (kmol/m ³)	—	57.1801	57.1801
Thermal Expansion Coefficient (1/K)	—	0.00245895	0.00245895
Heat Capacity at Constant Pressure (kJ/(kmol K))	—	39.1253	39.1253
Heat Capacity at Constant Volume (kJ/(kmol K))	—	24.7050	24.7050
Joule–Thomson Coefficient (K/bar)	—	-0.035715	-0.035715
Speed of Sound (m/s)	—	689.968	689.968

10 Discussion

The results table shows that the spreadsheet calculator produced reasonable values for many of the secondary thermodynamic properties. In particular, the calculated heat capacities satisfy the expected thermodynamic relation $C_p > C_v$, and the predicted Joule–Thomson coefficient remains negative at the specified state, consistent with the NIST trend.

The NIST comparison indicates that the EOS-based predictions are in the same general range as the reference data, although there are clear quantitative deviations. The speed of sound predicted by the EOS-based calculations is noticeably lower than the NIST value. The density and residual-property-based values also differ from the NIST reference, which

suggests that the chosen constants and EOS form do not perfectly reproduce the real-fluid behavior of nitrogen at this very high pressure.

The Peng–Robinson, exponential EOS with theoretical constants, and exponential EOS with numerical constants gave nearly identical outputs in the completed spreadsheet. This suggests that either the calculator structure treated the EOS similarly under the chosen inputs, or the parameter implementation did not create enough separation between the three models at the state considered. Even so, the template was completed and the resulting values are organized in the form requested by the assignment.

11 Conclusion

This report completed the comparison of nitrogen properties at 10 °C and 830 bar using experimental data, the Peng–Robinson EOS, an exponential EOS with theoretical constants, an exponential EOS with numerical constants, and NIST reference values. The required table format was reproduced using the Excel calculator template. The final comparison shows that the EOS-based predictions are reasonable in trend but differ quantitatively from NIST values, especially for density, enthalpy, entropy, and speed of sound. The completed table still provides a useful basis for discussing the strengths and limitations of each equation of state.

12 Acknowledgment of AI Use

Artificial intelligence was used to help organize the assignment content and format the report into LaTeX for Overleaf. AI was also used to help interpret the Excel calculator layout and transfer the completed table into report form.

13 References

- ChatGPT, used to help organize the report and format the final document in LaTeX.